

NATIVE SOURCE. Power development; safety/PHY/PCB/bench qualification remain OPEN.

01_carrier_interface

File: 01_carrier_interface.kicad_sch

20_positive_rails

File: 20_positive_rails.kicad_sch

40_power_supervision

File: 40_power_supervision.kicad_sch

04_dac_0_3

File: 04_dac_0_3.kicad_sch

06_analog_outputs

File: 06_analog_outputs.kicad_sch

10_input_protection

File: 10_input_protection.kicad_sch

30_negative_reference

File: 30_negative_reference.kicad_sch

03_peripheral_boundaries

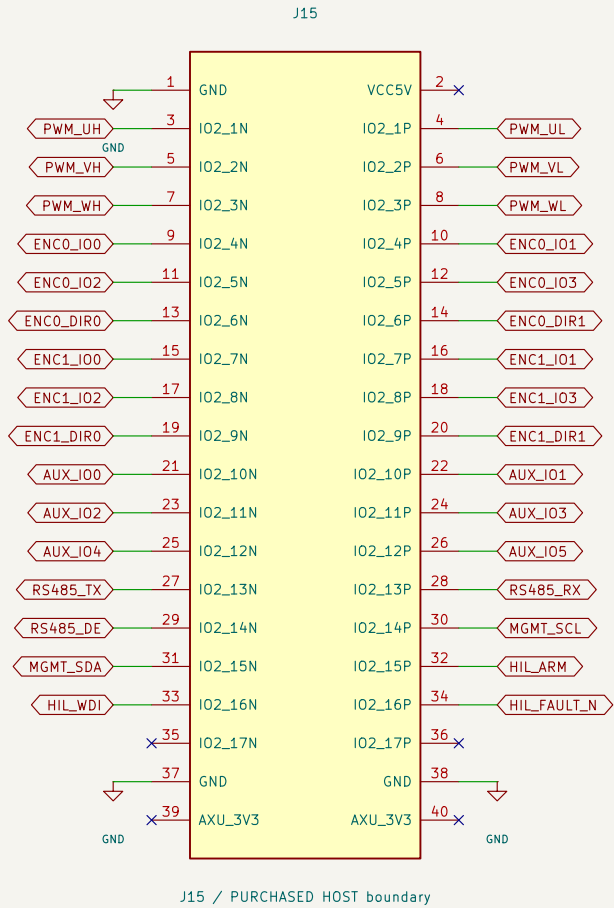
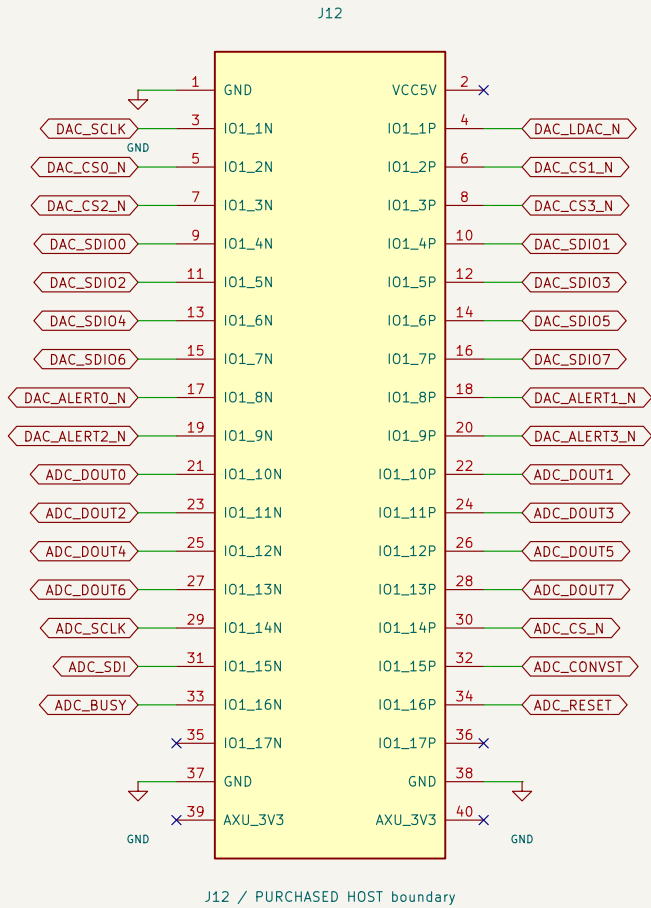
File: 03_peripheraL_boundaries.kicad_sch

05_dac_4_7

File: 05_dac_4_7.kicad_sch

41_power_status

File: 41_power_status.kicad_sch



Connector power contacts remain NC. 64 assigned GPIO; unused GPIO also NC. No secondary compute chip.

J12 and J15 are ORIGINAL AXU2CGB interfaces, not AXU2CGB-I or -E. No board-power backfeed through contacts.

Signal-pin backfeed, output enable and bank power sequencing STILL REQUIRE electrical review.

NON-RELEASE: physical power, safety, timing and package reviews remain open

magic-att/servoHIL

Sheet: /01_carrier_interface/

File: 01_carrier_interface.kicad_sch

Title: 01 carrier interface

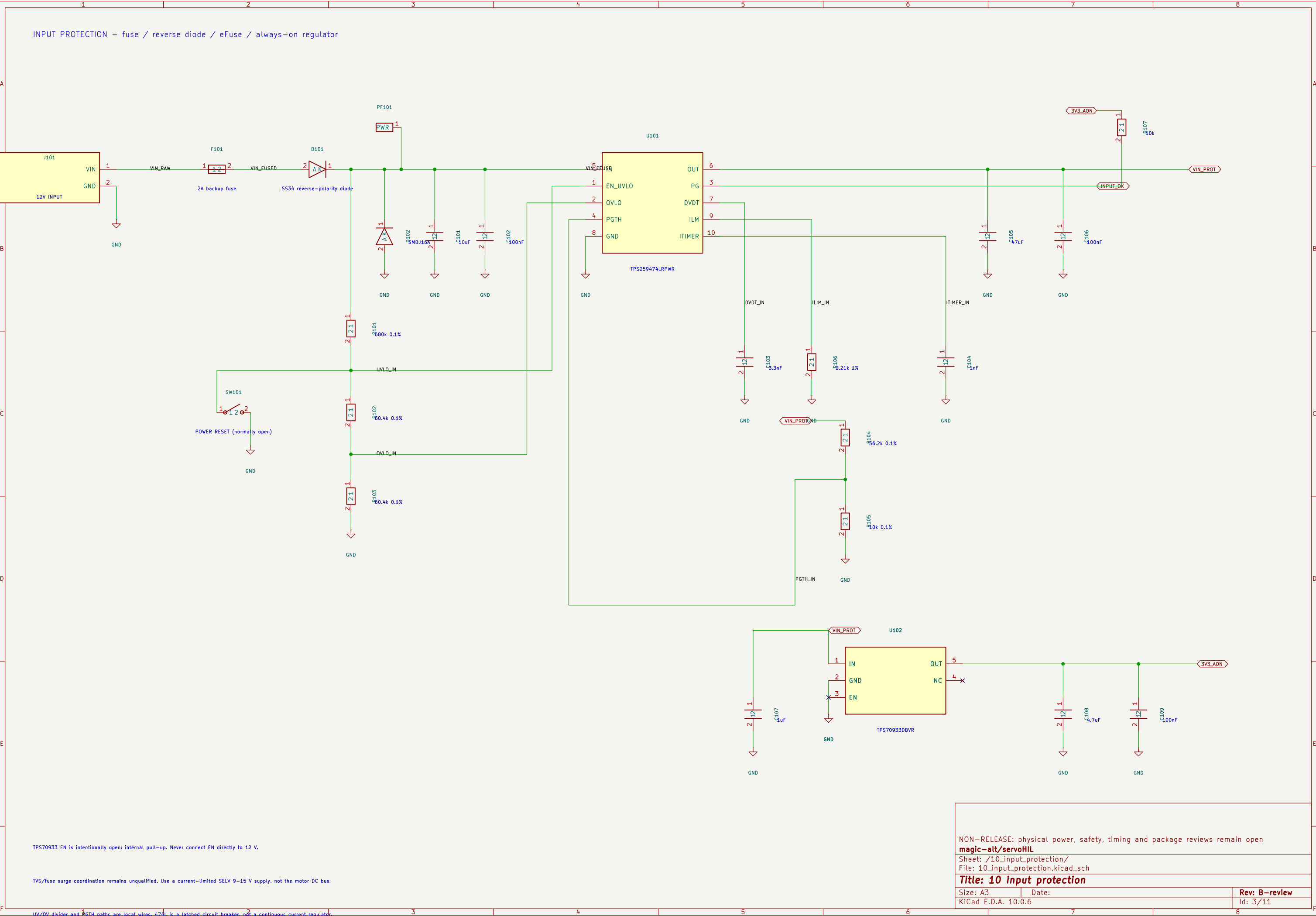
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Date:

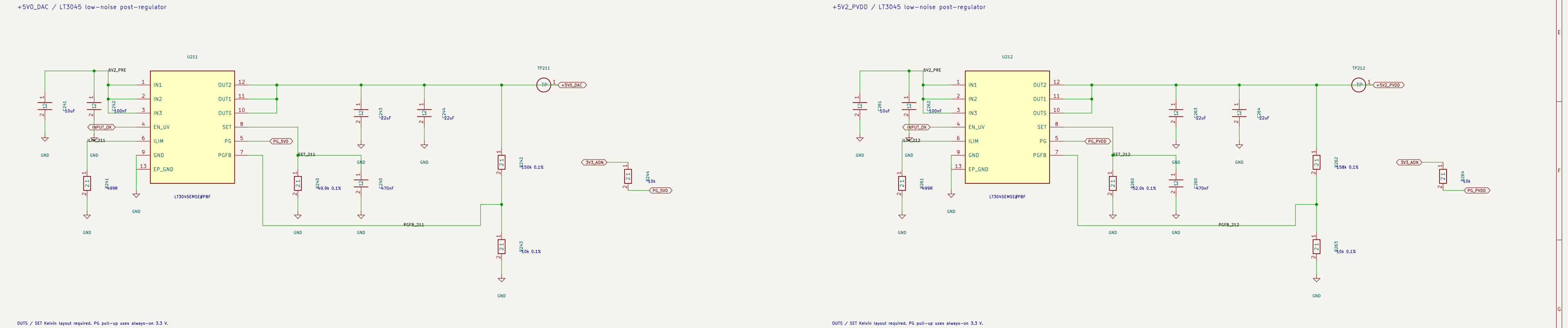
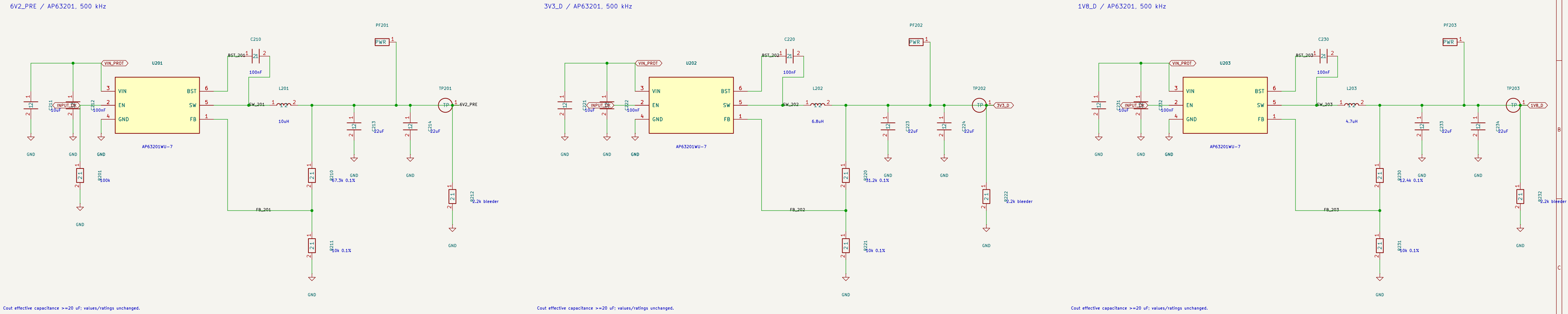
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POSITIVE RAILS – direct power path, local feedback, cross-sheet interfaces only



All parts, pin numbers, values and footprints match the previous native circuit. These drawings do not establish bench qualification.

No second FPGA. Output current, capacitor DC bias, startup, ripple, thermal and safety behavior remain hardware review gates.

NON-RELEASE: physical power, safety, timing and package reviews remain open

magic-alt/servoHIL

Sheet: /20_positive_rails/

File: 20_positive_rails.kicad_sch

Title: 20 positive rails

Size: A2

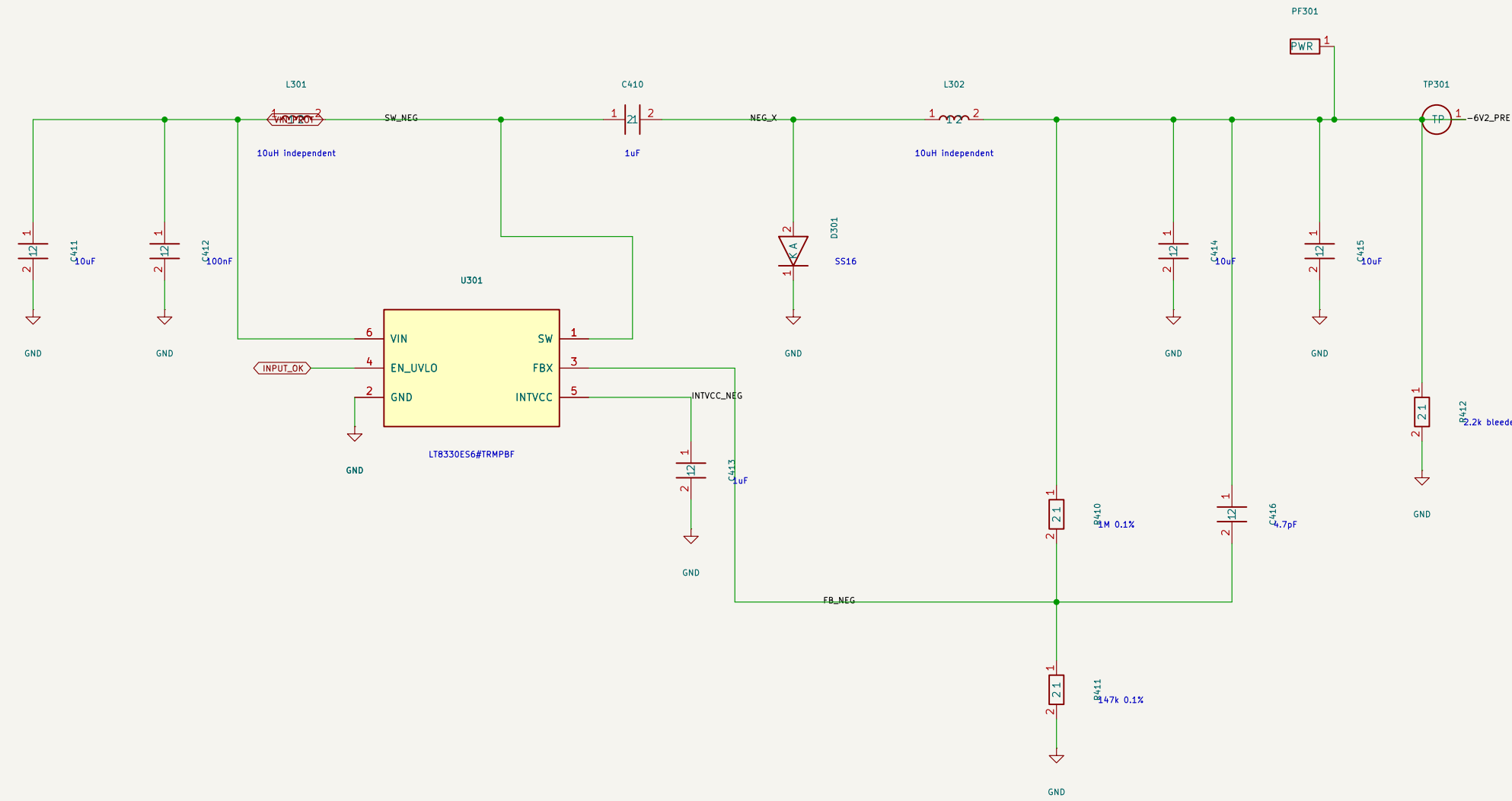
Date:

Rev: B-review

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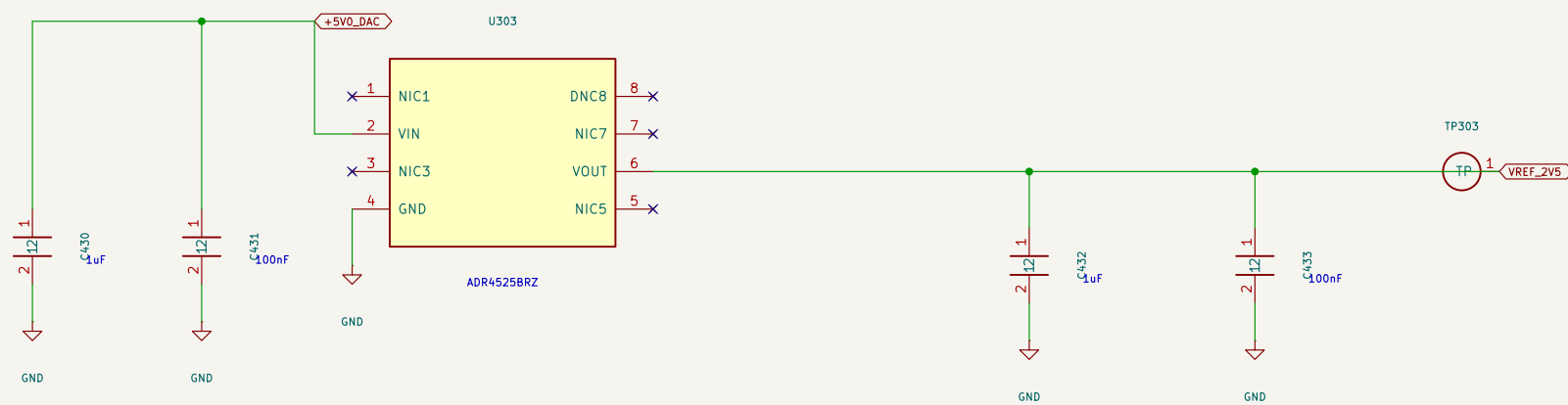
Id: 4/11

NEGATIVE RAIL / REFERENCE – wired Cuk converter, LT3094 and ADR4525



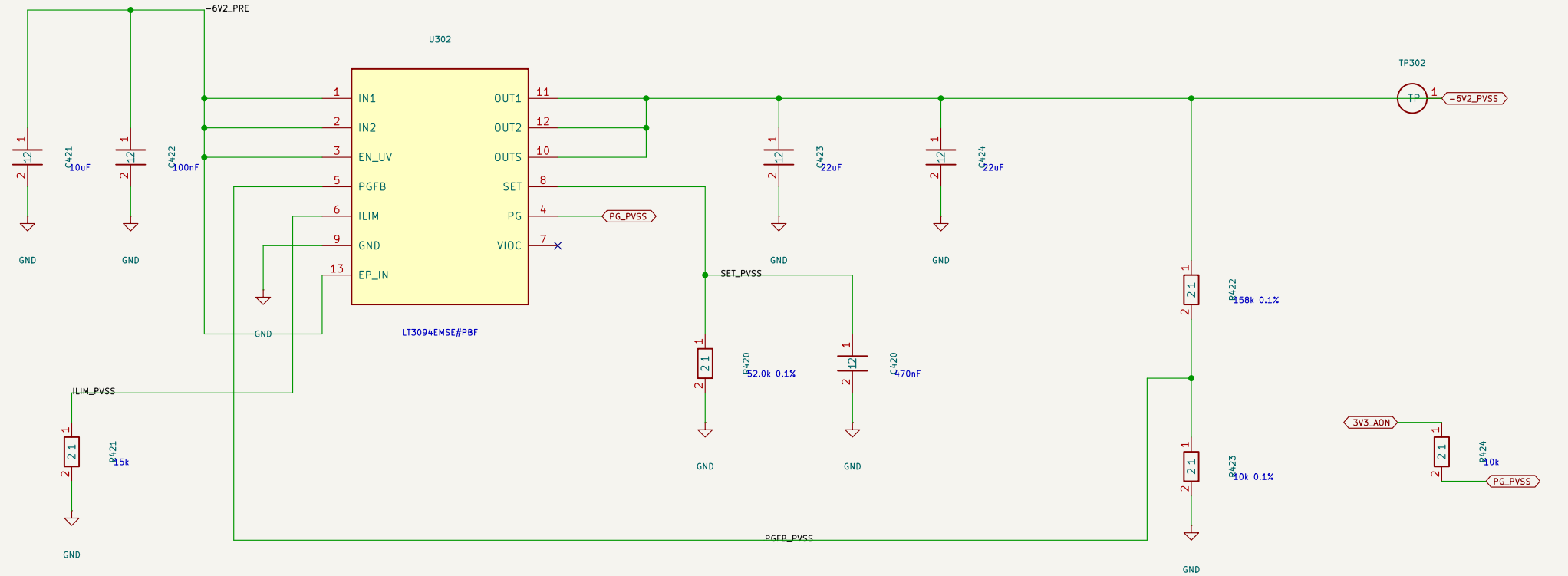
Cuk current path: VIN -> L301 -> C410 -> L302 -> negative output. D301 cathode is grounded

Independent inductors; transfer capacitor sees $V_{IN} + |V_{OUT}|$. Existing component values are preserved.

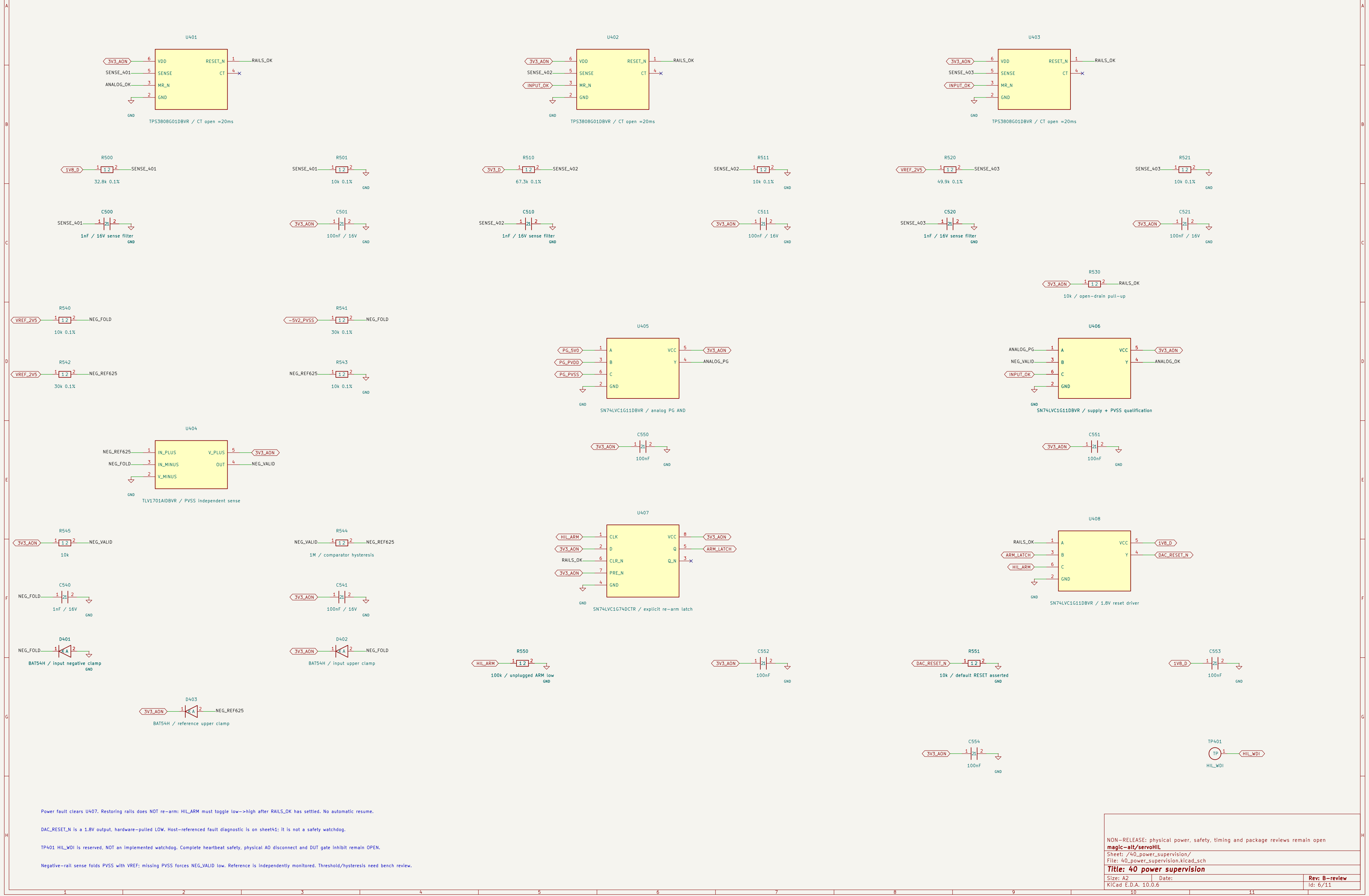


LT3094 EP13 / EN3 connect to negative IN. OUTS/SET require Kelvin layout. No startup/ripple/thermal PASS is implied

ADR4525: unused NIC/DNC pads remain NC. DAC external—reference configuration and actual fanout/load require validation.



POWER QUALIFICATION: monitored rails, negative-rail loss detection and latched DAC reset release



Power fault clears U407. Restoring rails does NOT re-arm: HIL_ARM must toggle low-->high after RAILS_OK has settled. No automatic resume.

DAC_RESET_N is a 1.8V output, hardware-pulled LOW. Host-referenced fault diagnostic is on sheet41; it is not a safety watchdog.

TP401 HIL_WDI is reserved, NOT an implemented watchdog. Complete heartbeat safety, physical AO disconnect and DUT gate Inhibit remain OPEN.

Negative-rail sense folds PVSS with VREF; missing PVSS forces NEG_VALID low. Reference is independently monitored. Threshold/hysteresis need bench review.

NON-RELEASE: physical power, safety, timing and package reviews remain open

magic-alt/servoHIL

Sheet: /40_power_supervision/

File: 40_power_supervision.kicad_sch

Title: 40 power supervision

Size: A2

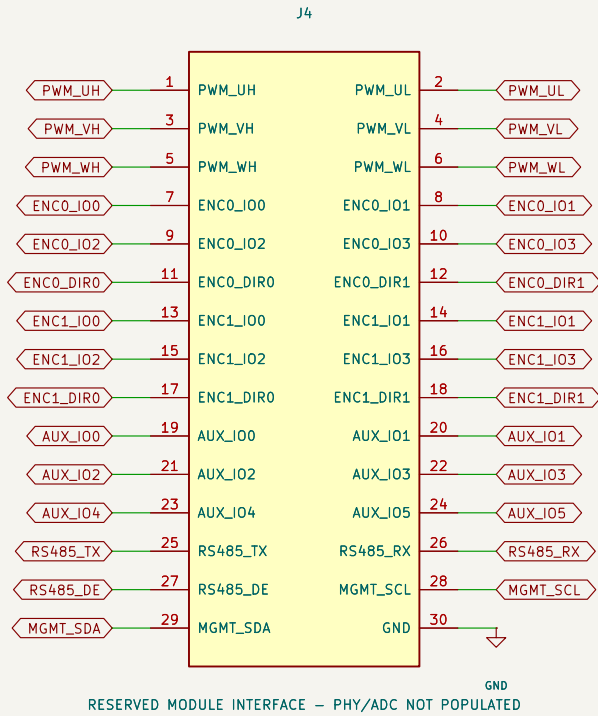
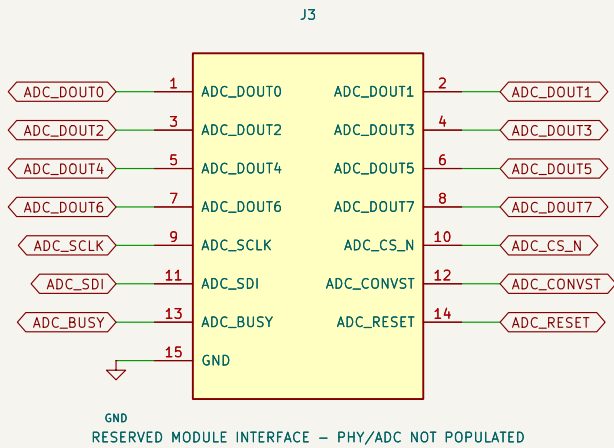
Date:

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Id: 6/11

03 – ADC and digital PHY integration interfaces (reserved)



Eight AI channels are an interface reservation. No 1 MSPS verification is implied; 1.8V serial timing must be closed.

Each encoder uses one selected profile; ABZ/SSI/BISS/SPI are NOT simultaneously independent ports.

NON–RELEASE: physical power, safety, timing and package reviews remain open

magic–alt/servoHIL

Sheet: /03_peripheral_boundaries/

File: 03_peripheral_boundaries.kicad_sch

Title: 03 peripheral boundaries

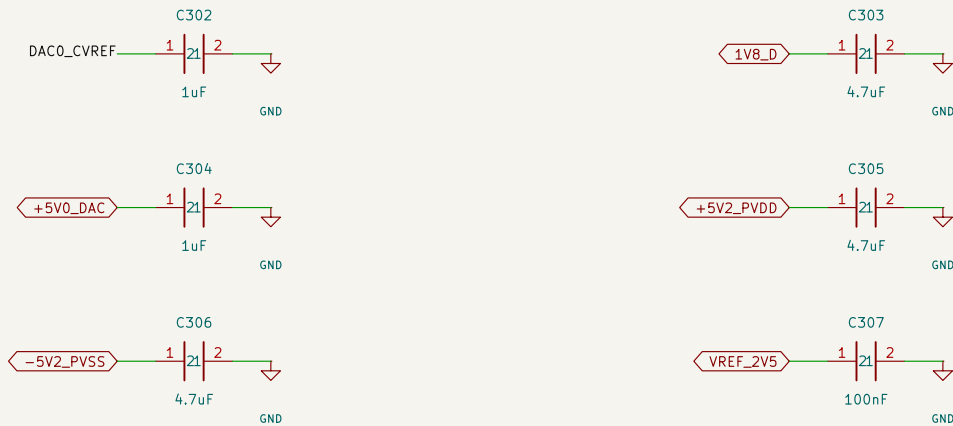
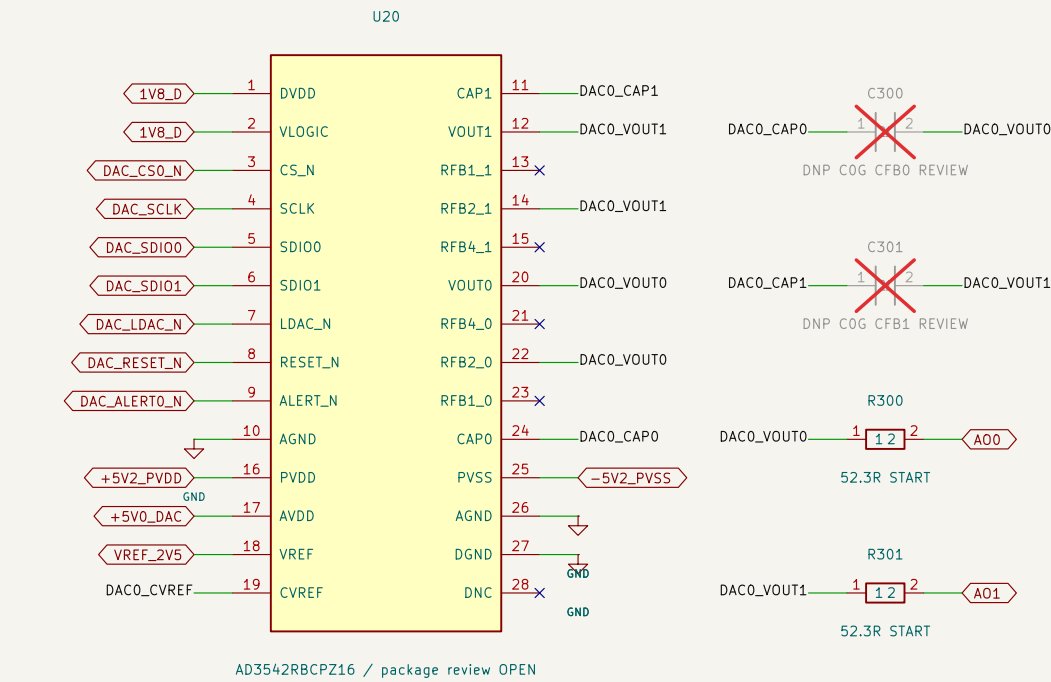
Size: A3

Date:

Rev: B–review

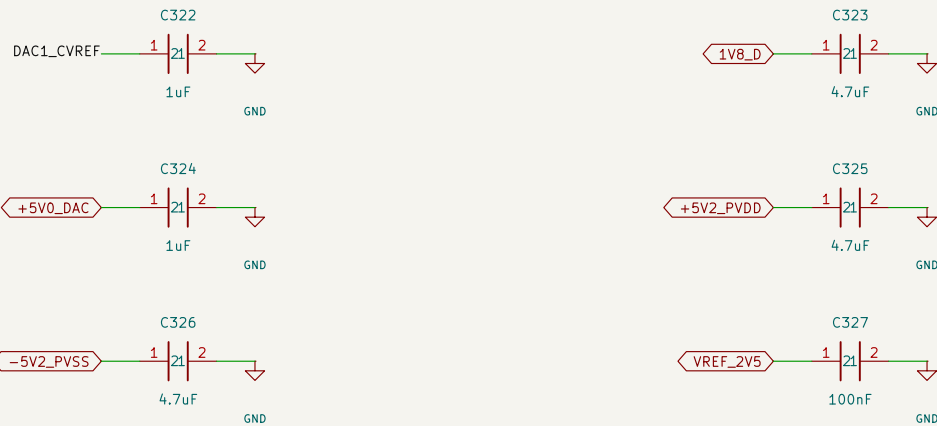
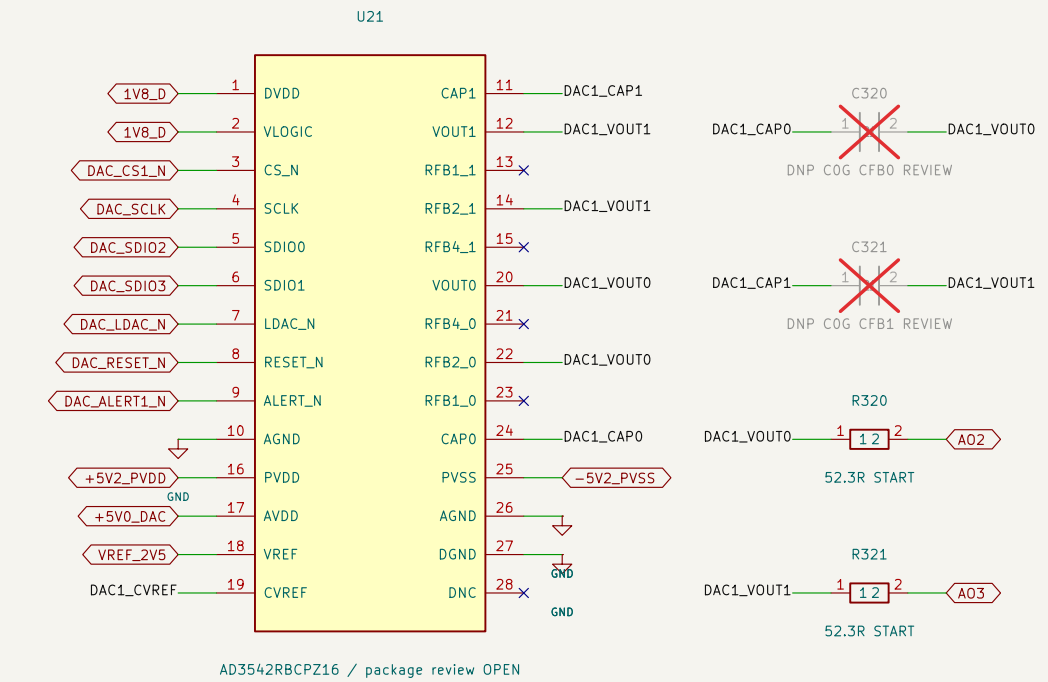
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Id: 7/11



RFB2 selects +/-5V capability; RFB1/RFB4/DNC are not connected, CFB and load stability are NOT approved.

External reference configuration and power–up register settings must be verified before enabling outputs.



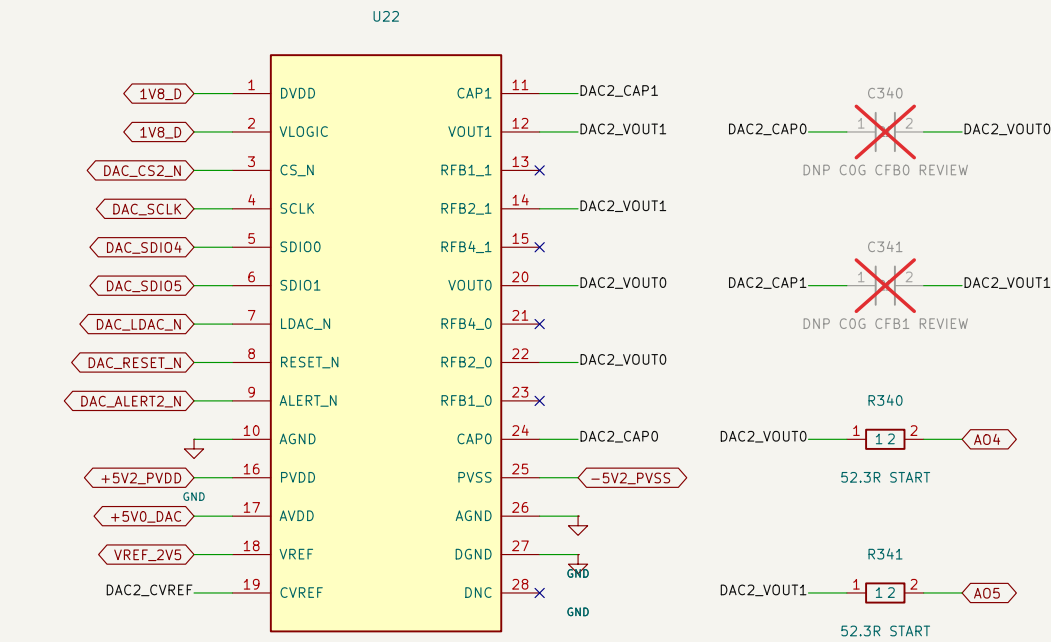
NON–RELEASE: physical power, safety, timing and package reviews remain open

magic–alt/servoHIL

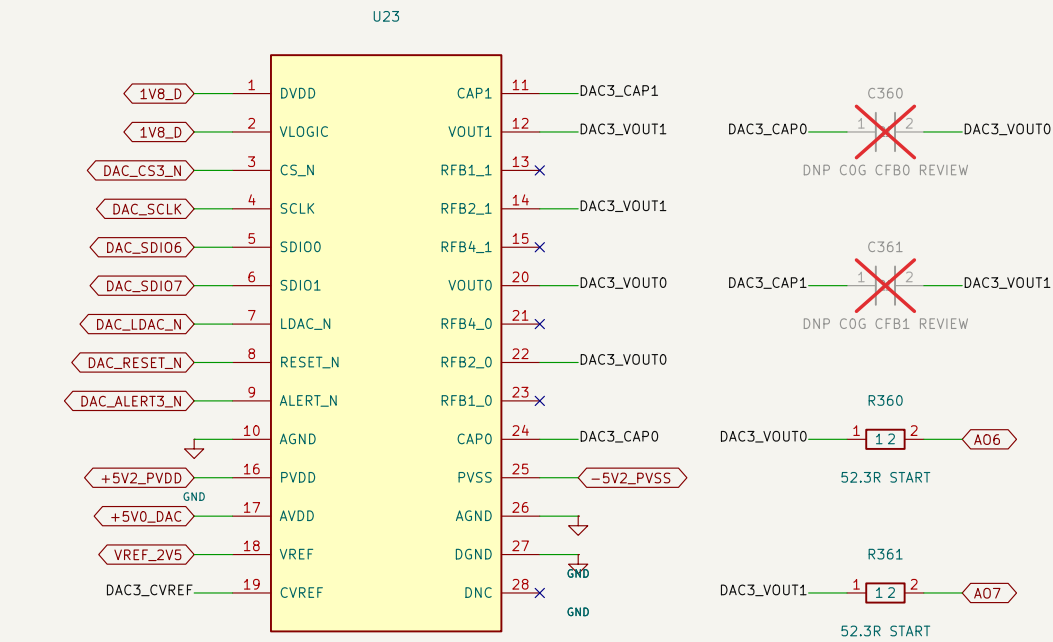
Sheet: /04_dac_0_3/
File: 04_dac_0_3.kicad_sch

Title: 04 dac 0 3

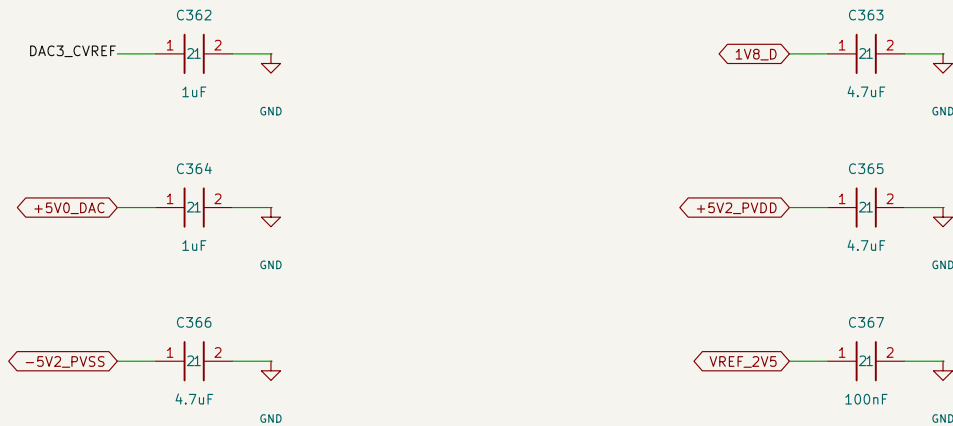
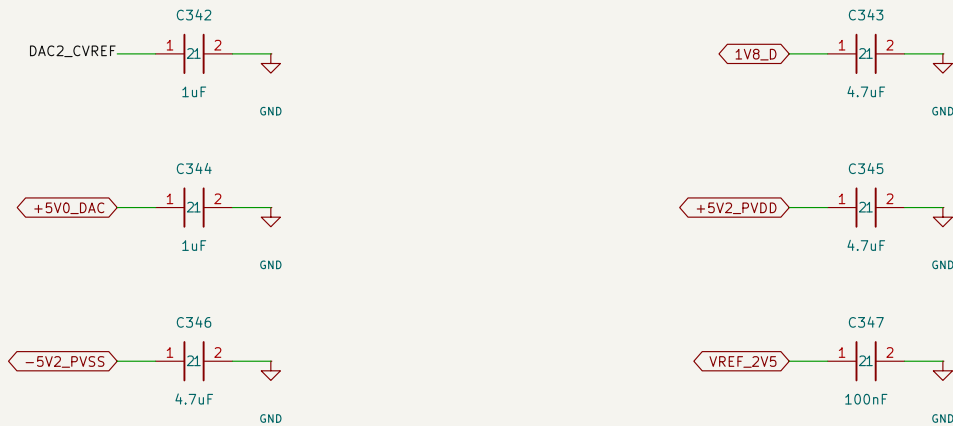
Size: A3	Date:	Rev: B–review
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AD3542RBCPZ16 / package review OPEN



AD3542RBCPZ16 / package review OPEN



RFB2 selects +/-5V capability; RFB1/RFB4/DNC are not connected, CFB and load stability are NOT approved.

External reference configuration and power–up register settings must be verified before enabling outputs.

NON–RELEASE: physical power, safety, timing and package reviews remain open

magic–alt/servoHIL

Sheet: /05_dac_4_7/

File: 05_dac_4_7.kicad_sch

Title: 05 dac 4 7

Size: A3

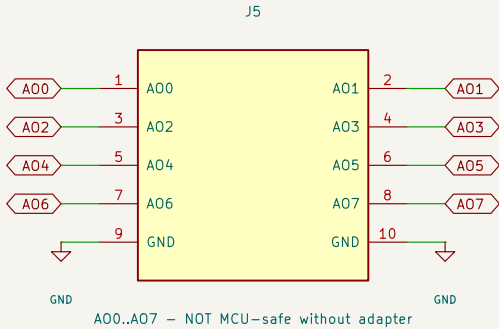
Date:

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06 – Eight analog outputs to reviewed DUT adapter



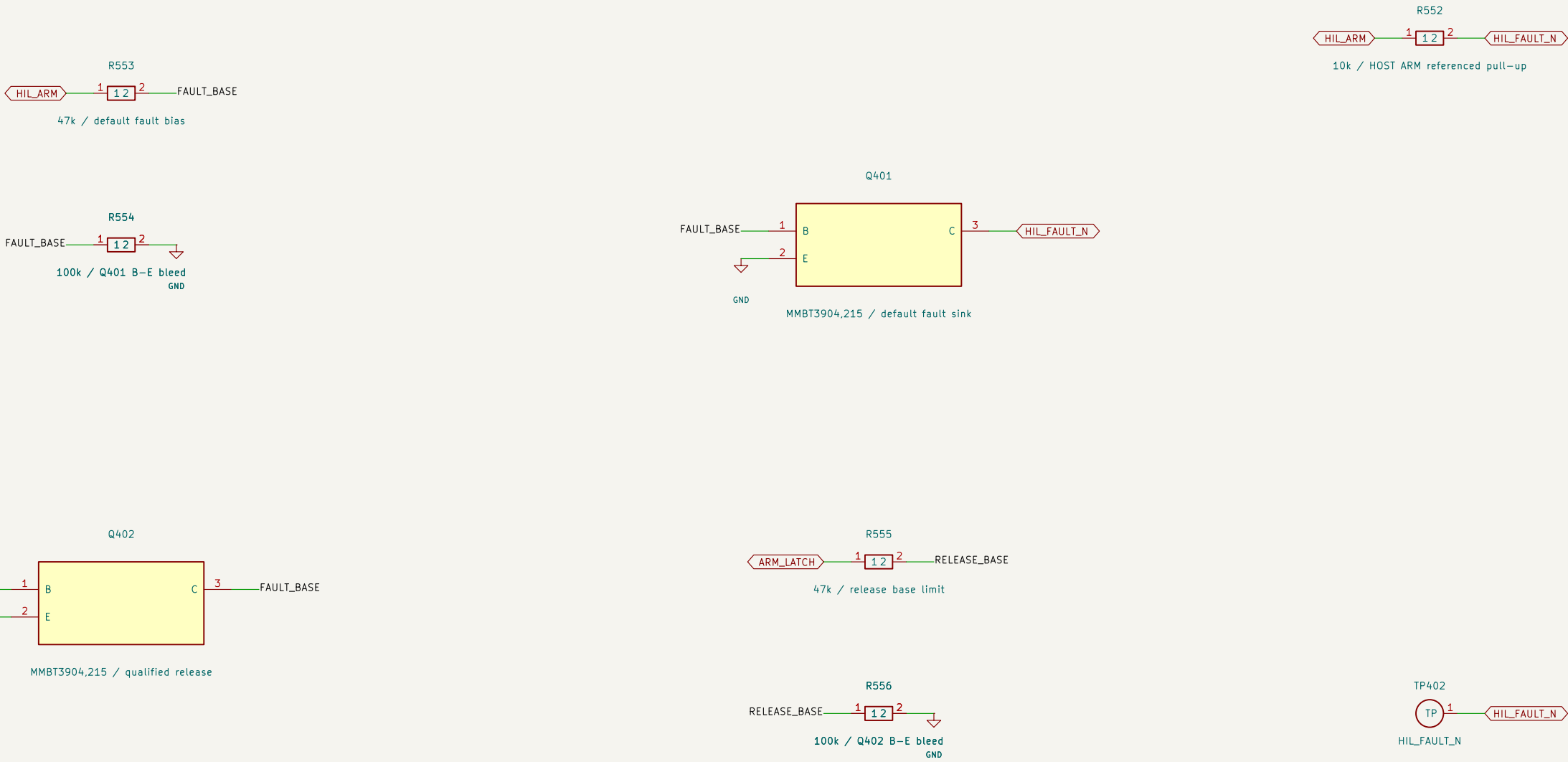
Default metadata: Ia, Ib, Ic, Vbus, torque, temperature, AUX0, AUX1. Electrical channels are equivalent.

DUT adapter supplies scaling/clamping and hardware inhibition. No direct +/-5V connection to MCU ADC.

NON-RELEASE: physical power, safety, timing and package reviews remain open
~~magic-alt/servoHIL~~
Sheet: /06_analog_outputs/
File: 06_analog_outputs.kicad_sch

Title: 06 analog outputs		
Size: A3	Date:	Rev: B-review
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HOST-REFERENCED POWER STATUS: no expansion-rail pull-up on HIL_FAULT_N



HIL_ARM is generated by the powered HOST, NOT an expansion rail. Q401 defaults ON while HOST requests ARM.

Only a power-qualified ARM_LATCH turns Q402 on and releases Q401. Board power lost -> Q402 off -> Q401 reports fault.

HIL_FAULT_N=HIGH means power-qualified AND armed; LOW also means disarmed. At HOST off, no expansion supply drives this pin high.

Nominal fault sink <0.33mA and HOST ARM total load <0.4mA. Check transient/partial-power behavior physically; not certified safety.

Other direct DAC/ADC pins STILL require power-off backfeed protection. This diagnostic alone does not authorize hot plugging.

NON-RELEASE: physical power, safety, timing and package reviews remain open

magic-alt/servoHIL

Sheet: /41_power_status/

File: 41_power_status.kicad_sch

Title: 41 power status

Size: A3

Date:

Rev: B-review

KiCad E.D.A. 10.0.6

Id: 11/11